

aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun2

Standard of care options — approved and guideline-endorsed strategies

Generated 2026-08-07

Standard of care options

The treatment strategies that are standard for this patient's situation, meaning a regulator approved them for a population that includes this patient or a major academic or clinical-society guideline carries them. This report runs alongside the targetable-feature ranking rather than in place of it, and it does not narrow what that ranking surfaces.

Assessed: 15 standard options. **Consider now:** 7. **Behind an open gate:** 0. **Already received:** 2.

Fifteen standard options were assessed for this relapsed/refractory AML-MR, and seven can be weighed now. The headline is uncomfortable and worth stating plainly: for TP53-mutated poor-risk AML, NCCN AML v3.2026 lists clinical trial and nothing else, and no approved regimen performs well in this disease state. That makes trial enrollment the guideline-preferred choice, not a consolation. The one on-label drug is gemtuzumab ozogamicin, approved for relapsed/refractory CD33-positive AML, and her blasts are CD33-positive; its own best randomised evidence shows no survival benefit in her adverse karyotype, so it is standard by approval with a modest expectation. Hydroxyurea for a peripheral blast count that moved 58% to 82% in a fortnight, concurrent palliative-care involvement while fitness and goals are still undocumented, high-dose cytarabine salvage as a cytoreductive bridge, an NCCN-listed hypomethylating agent, and radiotherapy to the thigh myeloid sarcoma, which guidelines cover only thinly, complete the near-term list.

Two standard options are already spent: the first sibling allograft, which held remission for about six years, and the FLAG-IDA plus venetoclax reinduction the disease progressed through. That venetoclax exposure also closes several venetoclax-dependent routes elsewhere in the case.

The curative destination, a second allograft with sibling DLI as the other cellular route, carries no NCCN category; the ELN 2022 panel and the EBMT 2024 DLI recommendations are the graded texts, and both sit behind cytoreduction and an organ-function workup that has not been done. Revumenib, ziftomenib, and CPX-351 are assessed and set aside on genotype or line, and the FLT3- and IDH-wild-type results close those inhibitor labels.

This page is additive. It records the standard options and does not narrow, reorder, or filter the experimental options ranked elsewhere on this case page.

Options to consider now

Standard options this patient is eligible for, with an endorsement that covers their situation, and no unresolved gate in front of them.

Clinical trial enrollment

Priority: essential **Modality:** Other **Line:** Salvage **Intent:** curative intent

What makes it standard.

- NCCN Preferred; sole listed option for TP53-mutated poor-risk AML (covers this patient · NCCN AML v3.2026 (2025-11-24))
 - covers: Relapsed/refractory AML: clinical trial enrollment is the preferred route. For poor-risk AML with TP53 mutation or del(17p), the guideline lists clinical trial and nothing else.

- other_society ELN 2022: expert-panel recommendation (covers this patient · ELN 2022)
 - covers: Relapsed/refractory AML after allogeneic HCT: donor lymphocyte infusion and second allogeneic HCT are treatment options for eligible patients; active disease and shorter remission predict worse outcomes, and patients with active disease should be considered for clinical trials.

Fit to this patient. Likely eligible ECOG 1, curative intent, and a stated acceptance of trials including travel are on file. Eligibility for any specific protocol still turns on that protocol's criteria and on the pending HLA and HA-1/HA-2 typing plus the TP53 and organ-function workup.

NCCN AML v3.2026 answers this disease state directly: for poor-risk AML with TP53 mutation or del(17p) it lists clinical trial and nothing else, and for relapsed/refractory disease generally it makes trial enrollment the preferred route. The honest reading is that no approved regimen performs well enough here to displace it, and this report says so rather than manufacturing a standard option. ECOG 1, curative intent, and willingness to travel keep her a realistic candidate; which protocols exist for her turns on the pending workup.

Toxicities that would change the decision. Protocol-specific; varies by study

Alongside the targeted options. Can run alongside the targeted options. Every live option in the experimental ranking reaches her through a trial or a protocolised route, from the rank-1 gating workup through CLAG-GO, the second allograft, HA-1 TCR-T, and DLI. This row is the delivery mechanism for those options, not a competitor to them.

Last verified. 2026-08-07

References. guideline:NCCN AML v3.2026; pmid:35797463

Gemtuzumab ozogamicin, single agent (CD33 antibody-drug conjugate)

Priority: high **Modality:** Targeted therapy **Line:** Salvage **Intent:** disease-control intent

What makes it standard.

- FDA On-label approval (single agent) (covers this patient · FDA approval 2017-09-01)
 - covers: Relapsed or refractory CD33-positive acute myeloid leukemia in adults and in pediatric patients 2 years and older; single-agent fractionated dosing 3 mg/m² on days 1, 4, and 7.
- NCCN Category 2A (CD33-positive R/R AML) (covers this patient · NCCN AML v3.2026 (2025-11-24))
 - covers: Relapsed/refractory CD33-positive AML: single-agent gemtuzumab ozogamicin. The drug is absent from the poor-risk induction options while present for favourable and intermediate risk.

Fit to this patient. Conditionally eligible CD33 positivity by flow meets the label. No bilirubin or transaminases are documented, and the boxed hepatotoxicity warning matters more than usual in a patient headed toward a second conditioning regimen.

- blocker: No baseline liver function on file; boxed warning for hepatotoxicity including veno-occlusive disease, and the risk stacks with a planned second transplant
- blocker: Best-quality randomised evidence (Hills 2014 IPD meta-analysis) shows no survival benefit in adverse cytogenetics, her exact karyotype group

Biomarker gate. CD33 expression CD33-positive by flow cytometry (label); quantitative antigen density not required by label but requested in the case workup (threshold met)

Gemtuzumab ozogamicin is the one drug on this page approved for exactly her situation: single-agent use in relapsed or refractory CD33-positive AML in adults, and her blasts are CD33-positive by flow. The expectation has to be priced honestly, because the fractionated schedule gave 26% CR in first relapse with a clean pre-transplant hepatic record, yet ALFA-0701 excluded secondary AML, which is her category, and the 3,325-patient meta-analysis found no survival benefit in adverse cytogenetics (OR 0.99, 95% CI 0.83-1.18). Baseline liver labs come first, since the veno-occlusive disease warning stacks against a planned second conditioning.

Key evidence.

- MyloFrance-1 (Taksin 2007) · single-arm phase 2, fractionated single-agent GO · n=57 · CR 26% (15/57); no grade 3-4 hepatotoxicity and no veno-occlusive disease reported; 7 patients transplanted afterwards; median neutropenia 23 days
- Hills 2014 IPD meta-analysis · individual-patient-data meta-analysis of GO added to induction · n=3325 · No survival benefit in adverse cytogenetics: OR 0.99 (95% CI 0.83-1.18, p=0.9)

Toxicities that would change the decision. Veno-occlusive disease / sinusoidal obstruction syndrome (boxed warning), the wrong risk to invite just before second-transplant conditioning; Prolonged myelosuppression in a six-year-old graft; median neutropenia 23 days in the fractionated series

Alongside the targeted options. Competes for the same line of therapy. Same CD33 payload as the rank-2 CLAG-GO regimen, which stays in the experimental table because the combination is trial-only with no published efficacy in refractory AML. Single-agent GO is the on-label route if the NCT04050280 slot or intensive fitness fails; giving it alone first spends the hepatic and CD33 exposure the rank-2 regimen plans to use.

Last verified. 2026-08-07

References. fda:Mylotarg; guideline:NCCN AML v3.2026; pmid:17051246; pmid:25008258

Hydroxyurea for blast-count control (with leukapheresis if needed)

Priority: high **Modality:** Supportive care **Line:** Any line **Intent:** disease-control intent

What makes it standard.

- NCCN Supportive-care recommendation (hyperleukocytosis / blast-count control) (covers this patient · NCCN AML v3.2026 (2025-11-24))
 - covers: AML with hyperleukocytosis or rapidly rising blast counts: hydroxyurea, with leukapheresis where indicated, as temporising cyto-reduction while definitive therapy is arranged.

Fit to this patient. Eligible Oral, titratable, and deliverable regardless of the undocumented organ-function numbers; peripheral blasts moving 58% to 82% inside a fortnight are the exact situation the recommendation covers.

Hydroxyurea with or without leukapheresis is the NCCN-carried supportive measure for blast-count control, and peripheral blasts that moved from 58% to 82% inside a fortnight make it immediately relevant. It changes counts rather than biology and buys days to weeks, not remission. Several downstream protocols also carry count ceilings that hydroxyurea is the usual way to reach.

Toxicities that would change the decision. Cytopenias and mucocutaneous effects; no disease-modifying activity

Alongside the targeted options. Comes before the targeted options. Several experimental-table protocols carry count ceilings that hydroxyurea is the usual way to reach: CBX-250 wants white cells under 25,000 and the 211At-BC8-B10 protocol wants circulating blasts under 10,000/mm³. It also buys days ahead of the rank-2 cytoreduction without spending any drug class the plan needs later.

Last verified. 2026-08-07

References. guideline:NCCN AML v3.2026

Concurrent specialty palliative care and goals-of-care assessment

Priority: high **Modality:** Supportive care **Line:** Not line-indexed **Intent:** symptom palliation

What makes it standard.

- ASCO Recommended (evidence-based, ASCO guideline update 2024) (covers this patient · ASCO 2024 (J Clin Oncol 42:2336-2357))
 - covers: Patients with advanced cancer, explicitly including hematologic malignancies and patients on early-phase trials, should be referred to specialty palliative care early and concurrent with active treatment.

Fit to this patient. Eligible Applies to any patient with advanced disease; nothing in the profile blocks it, and the undocumented fitness and goals-of-care assessment is an explicit open item in her own record.

ASCO's 2024 update tells oncology teams to involve specialty palliative care early and alongside active treatment for advanced cancer, hematologic malignancies included, and that holds with curative intent on the table. In this case the fitness and goals-of-care assessment is an explicit open item, and her own preferences file asks that the efficacy weighting be revisited once fitness is known. Concurrent palliative involvement is how that conversation happens without taking anything off the table.

Toxicities that would change the decision. None; concurrent palliative care does not displace disease-directed therapy

Alongside the targeted options. Can run alongside the targeted options. The fitness and goals-of-care assessment this brings is the same open item the rank-1 workup names, and her preferences file asks that the 0.75 efficacy weighting be revisited once fitness is known. It informs the weighting of every option in both tables without changing any option's eligibility.

Last verified. 2026-08-07

References. pmid:38748941; doi:10.1200/JCO.24.00542

Hypomethylating agent, lower intensity (azacitidine or decitabine)

Priority: medium **Modality:** Systemic chemotherapy **Line:** Salvage **Intent:** disease-control intent

What makes it standard.

- NCCN Category 2A (less-intensive therapy) (covers this patient · NCCN AML v3.2026 (2025-11-24))
 - covers: Relapsed/refractory AML, less-intensive therapy: azacitidine or decitabine.

Fit to this patient. Eligible Deliverable without the organ-function documentation an intensive regimen needs; nothing in the profile blocks a hypomethylating agent.

Azacitidine or decitabine is the NCCN-listed lower-intensity option for relapsed/refractory AML, and it stays deliverable if fitness closes the intensive route. The honest numbers are poor: CR 11% and median OS 6.7 months (95% CI 6.1-7.3) in the largest reported R/R cohort, with more than 5% circulating blasts predicting worse, and hers are 82%. The approved azacitidine-venetoclax combination does not transfer either, because its label covers newly diagnosed patients unfit for intensive induction and she has already progressed on venetoclax.

Key evidence.

- Stahl 2018 (international R/R cohort) · retrospective multicentre cohort · n=655 · CR 11%, CR/CRi 16%; median OS 6.7 months (95% CI 6.1-7.3); more than 5% circulating blasts predicted worse response and survival

Toxicities that would change the decision. Cytopenias and infection over repeated cycles; Weeks-per-cycle kinetics against a marrow that moved 58% to 82% inside a fortnight

Alongside the targeted options. Competes for the same line of therapy. An HMA course spends the same weeks the rank-2 cyto-reduction would use, so choosing it is a tempo decision against an 85% blast marrow. Adding venetoclax to it would re-expose a drug she has already progressed through, which is the same objection the board logged against venetoclax-containing trial arms.

Last verified. 2026-08-07

References. guideline:NCCN AML v3.2026; pmid:29685952

Intensive salvage chemotherapy (high-dose cytarabine-based, e.g. MEC or FLAG-M)

Priority: medium **Modality:** Systemic chemotherapy **Line:** Salvage **Intent:** disease-control intent

What makes it standard.

- NCCN Category 2A (aggressive-therapy candidates) (partial population match · NCCN AML v3.2026 (2025-11-24))
 - covers: Relapsed/refractory AML in patients who are candidates for aggressive therapy: high-dose cytarabine-based salvage regimens (cladribine- or fludarabine-containing, MEC and peers), typically as a bridge to transplant.
- other_society ELN 2022: expert-panel recommendation (partial population match · ELN 2022)
 - covers: Relapsed/refractory AML: intensive salvage for fit patients aiming at remission and transplant; trials preferred where available.

Fit to this patient. Conditionally eligible The disease state fits the guideline entry, but she has already progressed through FLAG-IDA plus venetoclax, a member of the same high-dose cytarabine and purine-analog family, and no organ-function numbers exist to price another intensive cycle.

- blocker: Refractory to FLAG-IDA plus venetoclax; nucleoside and HiDAC cross-resistance caps the expected depth of response
- blocker: Organ function undocumented; prolonged aplasia in a six-year-old graft could end the transplant plan rather than enable it

High-dose cytarabine-based salvage (MEC, CLAG-M and peers) is what NCCN lists for relapsed/refractory AML in aggressive-therapy candidates, and it is the standard way to bring a marrow down toward transplant. She has already progressed through FLAG-IDA plus venetoclax, a regimen from the same family, so cross-resistance caps the expected depth and the guideline entry is a partial match to a patient one intensive failure further along. Its role here is cytoreduction toward the curative platform rather than a destination in itself.

Key evidence.

- NCCN-listed intensive salvage regimens · guideline-referenced practice; retrospective salvage series · Second-line intensive salvage CR rates generally 20-40%, lower after failure of a prior high-dose cytarabine regimen; used here as a cytoreductive bridge rather than a destination

Toxicities that would change the decision. Prolonged grade 4 cytopenias, febrile neutropenia, and infection; Cumulative anthracycline and cardiac exposure on top of prior idarubicin, with no ejection fraction on file

Alongside the targeted options. Competes for the same line of therapy. Rank-2 CLAG-GO is the trial-delivered member of this same regimen family; an off-protocol CLAG-M or MEC uses the same window, the same marrow reserve, and the same hepatic budget. The choice between them is a delivery-route question, not a different strategy.

Last verified. 2026-08-07

References. guideline:NCCN AML v3.2026; pmid:35797463

Radiotherapy to the right-thigh extramedullary myeloid sarcoma

Priority: medium **Modality:** Radiotherapy **Line:** Any line **Intent:** disease-control intent

What makes it standard.

- other_society ILROG consensus guideline (International Lymphoma Radiation Oncology Group) (covers this patient · ILROG 2018 (Int J Radiat Oncol Biol Phys 102:314-319))
 - covers: Extramedullary leukemia / chloroma: radiotherapy for symptomatic lesions, sites with inadequate response to systemic therapy, or local control; modest doses, commonly around 24 Gy.
- NCCN Brief mention (myeloid sarcoma management) (partial population match · NCCN AML v3.2026 (2025-11-24))
 - covers: Myeloid sarcoma: systemic therapy is primary; radiotherapy for symptomatic or bulky lesions or inadequate local control. Coverage of extramedullary AML in the pathway is brief.

Fit to this patient. Likely eligible A defined soft-tissue limb target on MRI is deliverable regardless of marrow burden or the undocumented organ-function numbers; it treats the thigh site only.

Guideline coverage of extramedullary AML is thin: NCCN mentions radiotherapy for myeloid sarcoma only in passing, and the most specific text is the 2018 ILROG consensus, which supports modest-dose radiotherapy, commonly around 24 Gy, for symptomatic lesions or sites systemic therapy fails to control. The right-thigh lesion is an adverse feature that marrow-directed conditioning would not treat, so the local plan deserves its own line in the sequence. Radiotherapy to a limb site can run alongside systemic therapy without spending any systemic option.

Toxicities that would change the decision. Local skin and soft-tissue reaction, fatigue; No effect on the dominant marrow disease

Alongside the targeted options. Can run alongside the targeted options. Marrow-directed conditioning does not treat the thigh lesion, a gap the board named against the rank-3 transplant plan, and whether a CD33-directed regimen reaches it is an open question in the rank-2 row (paired series detect CD33 in about 79% of lesions). Radiotherapy to a limb site can run in parallel without spending any systemic option.

Last verified. 2026-08-07

References. pmid:30191865; guideline:NCCN AML v3.2026

Options at the next decision point

Standard for a line this patient has not reached yet, or for a disease state that has not arrived. Listed now so the sequence is visible in advance.

Second allogeneic hematopoietic cell transplant (HCT2), cytoreduce first

Priority: high **Modality:** Transplant **Line:** Salvage **Intent:** curative intent

What makes it standard.

- other_society ELN 2022: expert-panel recommendation (covers this patient · ELN 2022)
 - covers: Relapsed/refractory AML after allogeneic HCT: donor lymphocyte infusion and second allogeneic HCT are treatment options for eligible patients; active disease and shorter remission predict worse outcomes, and patients with active disease should be considered for clinical trials.

Fit to this patient. Conditionally eligible The six-year first remission and the related-donor first graft are favourable on the registry axes, but no cardiac, renal, or hepatic numbers exist to choose a conditioning intensity, the marrow stands at roughly 85% blasts, and the sibling's consent to a second collection is unconfirmed.

- blocker: No organ-function baseline on file; conditioning intensity cannot be chosen without an ejection fraction and creatinine clearance
- blocker: Active disease at ~85% marrow blasts; blasts under 27% carry HR 0.53 (95% CI 0.40-0.72) for survival in the registry model, and the worst band sits at 4% +/- 2% at two years
- blocker: Sibling donor consent for a second collection not yet documented

A second allograft from the sibling is the curative-intent destination the case plan is built around, and the ELN 2022 panel keeps second transplant and DLI in play for eligible post-transplant relapses. NCCN attaches no category to a second allograft after post-transplant relapse, the pathway assumes a first transplant, so this is uncovered rather than contraindicated, and the registry evidence behind it is a prognostic model rather than a treatment effect. The stakes are curative, which makes a sequencing error expensive: taking her to conditioning at roughly 85% blasts sits in the worst registry band, so cyto-reduction and an organ-function baseline come first.

Key evidence.

- EBMT registry (Ruutu 2015) · retrospective registry analysis · n=2632 · 5-year relapse-free survival 15% across 2,632 second allografts
- Christopeit 2013 (German registry) · retrospective multicentre registry cohort · n=179 · 2-year OS 37% +/- 6% when the first graft came from a related donor versus 16% +/- 4% after an unrelated first graft

Toxicities that would change the decision. Non-relapse mortality from a second conditioning regimen and graft-versus-host disease; no registry row tabulates it for her fitness band; Prolonged cytopenias and infection

Alongside the targeted options. Comes before the targeted options. The same option holds rank 3 in the experimental table as a user-stated feature; the tables are additive, so it appears in both. It is the platform on which the rank-4 HA-1 TCR-T consolidation would be delivered, it is sequenced after the rank-2 cyto reduction, and it is the alternative to the rank-5 DLI at the same fork.

Last verified. 2026-08-07

References. pmid:35797463; pmid:26367221; pmid:23918951

Donor lymphocyte infusion (DLI) from the HLA-identical sibling, after cyto reduction

Priority: medium **Modality:** Cellular therapy **Line:** Salvage **Intent:** disease-control intent

What makes it standard.

- other_society EBMT best-practice recommendation (Practice Harmonisation and Guidelines Committee with the Cellular Therapy and Immunobiology Working Party) (partial population match · EBMT 2024 (Lancet Haematol 11:e448-e458))
 - covers: Relapse of haematological malignancy after allogeneic HCT: therapeutic DLI, sibling starting dose 1×10^7 CD3+ cells/kg with escalation; framed mainly for MRD-level or cyto reduced disease rather than high-burden overt relapse.
- other_society ELN 2022: expert-panel recommendation (covers this patient · ELN 2022)
 - covers: Relapsed/refractory AML after allogeneic HCT: donor lymphocyte infusion and second allogeneic HCT are treatment options for eligible patients; active disease and shorter remission predict worse outcomes, and patients with active disease should be considered for clinical trials.

Fit to this patient. Conditionally eligible The sibling donor exists and a late relapse favours response, but DLI into a marrow at roughly 85% blasts sits in the worst registry band, the relapse pattern reads as graft-versus-leukemia escape, and the sibling's consent to a fresh collection is not yet documented.

- blocker: Active disease at ~85% blasts; DLI given in aplasia or active disease carried 2-year OS 15% +/- 3% versus 56% +/- 10% in remission or with favourable karyotype
- blocker: Grade III-IV GVHD after DLI would close the door on HCT2 and on post-transplant cellular consolidation
- blocker: Relapse pattern consistent with graft-versus-leukemia escape, which tempers re-infusing the same repertoire

The EBMT 2024 best-practice recommendations are the only graded guideline text for post-transplant relapse, and they set therapeutic sibling DLI at a starting dose of 1×10^7 CD3+ cells/kg, written mainly for low-burden or cyto reduced disease; NCCN carries no DLI recommendation at all, and that silence should not be read as endorsement. Schmid's registry puts DLI given with active disease at 15% +/- 3% two-year survival against 56% +/- 10% in remission, and she stands at roughly 85% blasts today. It stays live as the lower-intensity graft-versus-leukemia route after cyto reduction, particularly if the fitness workup rules out a second conditioning regimen.

Key evidence.

- Schmid 2007 (EBMT registry) · retrospective registry comparison, DLI versus no DLI · n=399 · 2-year OS 21%

+/- 3% with DLI versus 9% +/- 2% without (p=0.04); within the DLI group, 15% +/- 3% with active disease versus 56% +/- 10% in remission

Toxicities that would change the decision. Acute and chronic graft-versus-host disease, which is both the mechanism and the toxicity; Marrow aplasia after infusion into a heavily pretreated graft

Alongside the targeted options. Competes for the same line of therapy. The same option holds rank 5 in the experimental table as a user-stated feature; the tables are additive, so it appears in both. It sits at the same post-cytoreduction fork as HCT2, shares the sibling donor and collection window the rank-4 HA-1 TCR-T needs, and severe GVHD after it would foreclose both.

Last verified. 2026-08-07

References. pmid:38796194; pmid:35797463; pmid:17909197

Best supportive care and hospice referral

Priority: low **Modality:** Supportive care **Line:** Not line-indexed **Intent:** symptom palliation

What makes it standard.

- NCCN Named management strategy (covers this patient · NCCN AML v3.2026 (2025-11-24))
 - covers: Relapsed/refractory AML: best supportive care, including transfusion support, infection management, and hospice, where disease-directed therapy is not deliverable or not wanted.
- other_society ELN 2022: expert-panel recommendation (covers this patient · ELN 2022)
 - covers: Relapsed/refractory AML after allogeneic HCT: donor lymphocyte infusion and second allogeneic HCT are treatment options for eligible patients; active disease and shorter remission predict worse outcomes, and patients with active disease should be considered for clinical trials.

Fit to this patient. Eligible Available to any patient; carried here as the standard alternative if the curative pathway proves undeliverable or is declined once fitness and goals are documented.

Best supportive care with transfusion support, infection management, and hospice referral is a named management strategy in relapsed/refractory AML, not a footnote. Curative intent is on file and fitness is not yet documented, so this sits at the next decision point rather than now. It becomes the primary path if the transplant and cellular plan proves undeliverable or unwanted once fitness and goals are on the table.

Toxicities that would change the decision. None from the strategy itself; the trade-off is forgoing disease-directed therapy

Alongside the targeted options. Independent of the targeted options. The path taken if the curative transplant and cellular pathway closes; it does not compete with any single ranked option.

Last verified. 2026-08-07

References. guideline:NCCN AML v3.2026; pmid:35797463

Already received

Carried so the report reads as a complete account of the standard options rather than a list that quietly omits what has been tried.

FLAG-IDA + venetoclax reinduction, given 2026

Priority: low **Modality:** Systemic chemotherapy **Line:** Salvage **Intent:** curative intent

What makes it standard.

- NCCN Category 2A (aggressive-therapy candidates) (covers this patient · NCCN AML v3.2026 (2025-11-24))
 - covers: Relapsed/refractory AML in aggressive-therapy candidates: fludarabine plus high-dose cytarabine and G-CSF with idarubicin, with or without venetoclax.

Fit to this patient. Already received

Prior exposure. Given as reinduction for the relapsed/refractory AML. Best response was progressive disease: the marrow rose to about 85% blasts, peripheral blasts moved from 58% to 82%, and an extramedullary right-thigh lesion is present. Refractoriness to this high-dose cytarabine, purine-analog, and venetoclax regimen is why neither venetoclax nor a FLAG-type backbone is offered back as a fresh option.

FLAG-IDA with venetoclax is a standard high-intensity salvage for fit relapsed/refractory AML, and it was the reinduction this patient received. It failed, with progression to an 85% blast marrow and extramedullary spread. It is recorded so the report reads honestly about why no venetoclax-containing or FLAG-type regimen is offered back as a fresh option.

Alongside the targeted options. May foreclose a targeted option. The venetoclax exposure already on record is what narrows several experimental-table routes: the tagraxofusp trial cohort needs a venetoclax washout, the rezatapopt mechanism wanted venetoclax as its killing partner, and the BMS-986497 triplet arm would re-expose it. The foreclosure has already happened; this row records it.

Last verified. 2026-08-07

References. guideline:NCCN AML v3.2026

First matched-sibling allogeneic transplant (HCT1), given ~2020

Priority: low **Modality:** Transplant **Line:** Not line-indexed **Intent:** curative intent

What makes it standard.

- NCCN Recommended curative modality (higher-risk MDS) (covers this patient · NCCN MDS (carried across versions; not re-verified at version level for this run))
 - covers: IPSS-R high / very-high risk MDS: allogeneic HCT is the potentially curative therapy for eligible patients.
- other_society ELN 2022: expert-panel recommendation (covers this patient · ELN 2022)
 - covers: Relapsed/refractory AML after allogeneic HCT: donor lymphocyte infusion and second allogeneic HCT are treatment options for eligible patients; active disease and shorter remission predict worse outcomes, and patients with active disease should be considered for clinical trials.

Fit to this patient. Already received

Prior exposure. HLA-identical sibling PBSC allograft around 2020 for RCMD, IPSS-R high/very-high MDS. Best response was complete remission, held for about 72 months. The relapse roughly six years later reads as graft-versus-leukemia escape rather than graft failure, which is part of why a second transplant and post-transplant cellular consolidation are under discussion.

The first matched-sibling allograft was the curative-intent standard for the high-risk MDS that preceded this AML, and it held a complete remission for about six years. It is logged so the record reads as a complete account rather than one that omits the transplant the current plan builds on. Its durable graft-versus-leukemia effect, and the eventual escape from it, shape the case for a second graft and for antigen-directed consolidation.

Last verified. 2026-08-07

References. guideline:NCCN MDS; pmid:35797463

Assessed and set aside

Screened against this patient and found not to apply. Listed for the audit trail.

Revumenib (menin inhibitor)

Priority: low **Modality:** Targeted therapy **Line:** Salvage **Intent:** disease-control intent

What makes it standard.

- FDA On-label approval (two indications) (different population · FDA approvals 2024-11 (KMT2A), 2025-10 (NPM1))
 - covers: Relapsed or refractory acute leukemia with a KMT2A translocation (adults and pediatric patients 1 year and older), and relapsed/refractory AML with a susceptible NPM1 mutation.
- NCCN Category 2A (KMT2A-rearranged or NPM1-mutant R/R AML) (different population · NCCN AML v3.2026 (2025-11-24))
 - covers: Relapsed/refractory AML with a KMT2A rearrangement or an NPM1 mutation.

Fit to this patient. Likely ineligible Her karyotype shows KMT2A amplification, a copy-number gain of four copies, not the KMT2A translocation the label and companion diagnostic require, and the disease is NPM1 wild-type.

- blocker: KMT2A amplification is a copy-number gain, not the translocation / rearrangement the approval requires
- blocker: NPM1 wild-type, so the second indication is also unmet

Biomarker gate. KMT2A rearrangement or NPM1 mutation KMT2A translocation/rearrangement OR susceptible NPM1 mutation (threshold not met)

Revumenib is approved for relapsed/refractory acute leukemia with a KMT2A translocation and for NPM1-mutant disease, and it is named here because her karyotype invites the misread: KMT2A amplification is a copy-number gain of four copies, not the rearrangement the label and its companion diagnostic require, and the disease is NPM1 wild-type. Menin inhibition depends on the dependency a KMT2A fusion or an NPM1 mutation creates, which amplification does not. Assessed and set aside on genotype.

Key evidence.

- AUGMENT-101 (KMT2A-rearranged R/R acute leukemia) · phase 1/2, single-arm · n=57 · CR+CRh 22.8% (95% CI 12.7-35.8) in efficacy-evaluable patients; ORR 63.2%

Toxicities that would change the decision. Differentiation syndrome and QTc prolongation, the menin-inhibitor class effects

Alongside the targeted options. Independent of the targeted options. Named to close the loop on the KMT2A-amplification versus KMT2A-rearrangement distinction, a common misread of this karyotype. It does not interact with the ranked options.

Last verified. 2026-08-07

References. fda:Revuforj; pmid:39121437; guideline:NCCN AML v3.2026

Ziftomenib (menin inhibitor)

Priority: low **Modality:** Targeted therapy **Line:** Salvage **Intent:** disease-control intent

What makes it standard.

- FDA On-label approval (different population · FDA approval 2025-11-13)
 - covers: Adults with relapsed or refractory AML with a susceptible NPM1 mutation.
- NCCN Category 2A (NPM1-mutant R/R AML) (different population · NCCN AML v3.2026 (2025-11-24))
 - covers: Relapsed/refractory AML with a susceptible NPM1 mutation.

Fit to this patient. Likely ineligible The disease is NPM1 wild-type, and the ziftomenib approval is restricted to susceptible NPM1 mutations. KMT2A amplification does not create the menin dependency the drug exploits.

- blocker: NPM1 wild-type; the approval is restricted to susceptible NPM1 mutations
- blocker: KMT2A amplification is a copy-number gain and does not confer menin-inhibitor sensitivity

Biomarker gate. NPM1 mutation susceptible NPM1 mutation (threshold not met)

Ziftomenib is the second approved menin inhibitor, but its label is confined to relapsed/refractory AML with a susceptible NPM1 mutation, and this disease is NPM1 wild-type. As with revumenib, the KMT2A amplification here is a copy-number gain rather than the fusion that would create menin-inhibitor sensitivity. The drug is approved but the target is absent, so it is assessed and set aside.

Toxicities that would change the decision. Differentiation syndrome, the menin-inhibitor class effect

Alongside the targeted options. Independent of the targeted options. Listed alongside revumenib so the report is explicit that the approved menin-inhibitor class does not apply to an NPM1-wild-type, KMT2A-amplified genotype. It does not interact with the ranked options.

Last verified. 2026-08-07

References. fda:Komzifti; pmid:41653248; guideline:NCCN AML v3.2026

CPX-351 (liposomal daunorubicin/cytarabine)

Priority: low **Modality:** Systemic chemotherapy **Line:** Not line-indexed **Intent:** curative intent

What makes it standard.

- FDA On-label approval (different population · FDA approval 2017-08)
 - covers: Newly diagnosed therapy-related AML or AML with myelodysplasia-related changes in adults and pediatric patients 1 year and older.
- NCCN Category 2A (newly diagnosed AML-MRC / t-AML) (different population · NCCN AML v3.2026 (2025-11-24))
 - covers: Newly diagnosed AML with myelodysplasia-related changes or therapy-related AML, induction in fit patients.

Fit to this patient. Likely ineligible The histology, AML-MR, matches the label, but the line does not: CPX-351 is approved for newly diagnosed disease and she is relapsed and refractory after a transplant. Prior idarubicin also adds cumulative anthracycline exposure.

- blocker: Approved for newly diagnosed AML-MRC / t-AML only; the patient is relapsed/refractory
- blocker: Prior anthracycline (idarubicin) exposure with no ejection fraction on file

CPX-351 is the standard intensive frontline choice for the AML-MR histology she carries, which is why a reader would look for it here. The approval and the NCCN listing are confined to newly diagnosed disease, and she is relapsed and refractory after a transplant, so the histology match does not open the drug at this line. It is assessed and set aside on line, not on target.

Last verified. 2026-08-07

References. fda:Vyxeos; guideline:NCCN AML v3.2026